# Calcurion Maths Worksheets

A-Level Mathematics

**Topic:** Statistics – Sampling & Data

**SOLUTIONS**

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**1.** Population = entire set of interest; sample = subset selected for study.
**2.** Census: data from every member; sample survey: data from a subset.
**3.** Advantage: complete, accurate; disadvantage: time-consuming, expensive, may be impractical.
**4.** Every member has equal chance; use random numbers from a sampling frame.
**5.** Select every $k$th item from a list after a random start.
**6.** Divide population into strata; sample randomly from each in proportion (or equal).
**7.** Interviewer selects to fill quotas by characteristics; non-random.
**8.** Year 12: $\frac{360}{600} \times 50 = 30$; Year 13: 20.
**9.** Generate 40 unique random numbers from 0001–2000; select corresponding customers.
**10.** List of all members of the population from which the sample is drawn.
**11.** Ensures representation of key groups; reduces sampling variability for those groups.
**12.** Non-random; interviewer bias; cannot calculate sampling error reliably.
**13.** Systematic (or opportunity if not from a list); limitation: may miss groups who exit less often.
**14.** Qualitative: non-numerical (e.g. colour); quantitative: numerical (e.g. height).
**15.** Discrete: countable (e.g. number of cars); continuous: measured (e.g. weight).
**16.** Arts 20, Science 35, Business 25.
**17.** Incomplete frame excludes some members $\Rightarrow$ under-coverage bias.
**18.** Stratified by department (proportional); ensures each department represented.
**19.** $k = 500/25 = 20$; random start 1–20; then every 20th name.
**20.** SRS: unbiased, allows inference, but needs full frame; opportunity: quick/cheap but biased.
**21.** Leading/biased wording ("excellent"); forces agreement.
**22.** Primary: collected by researcher (control); secondary: existing (cheaper, may not fit purpose).
**23.** Yes as estimate if sample random/representative; uncertainty remains (sampling error).
**24.** Define population; choose stratified/SRS; sampling frame (register); sample size; collect times.
**25.** Bias = systematic error; arises from non-random selection, incomplete frame, non-response, leading questions.